

Langfuse Implementation with RAG Search & Tracing

1. Tracing of Vector Search with Langfuse Tracing:

The screenshot shows the Langfuse web interface for a project named 'Hobby' under the organization 'Ajay's Organization'. The 'Tracing' section is active, displaying a list of traces. The table below shows the first few traces:

Start Time	Type	Name	Trace
2026-08-05 16:55:29	vector-search	search	search
2026-08-05 16:55:29	vector-search	search	search
2026-08-05 16:55:28	generate-embedding	search	search
2026-08-05 16:55:28	bm25-search	search	search
2026-08-05 16:55:28	bm25-search	search	search

The detailed view of the 'vector-search' trace shows the following information:

- Latency: 1.35s
- User ID: c5309c10-58f9-481b-89a2-c2ecddbe28a7
- Env: default
- Preview: Scores, Log View, Formatted, JSON, Beta
- Trace Data (JSON):

```
{
  "id": "t-42c249ee-7d88-4a21-86d9-c1a45c1920ac",
  "type": "SPAN",
  "name": "search (t-42c249)",
  "startTime": {
    "start": "2026-08-05 16:55:29",
    "end": "2026-08-05 16:55:30"
  },
  "endTime": null,
  "depth": 0
}
```

2.1 Implementation of BM25 Search & Hybrid Search Tracing through Vibe Coding and COT:

The screenshot shows the Vibe Coding interface for a project named 'rag-mongo-demo-v10'. The interface includes a file explorer on the left, a central workspace with a large 'X' icon, and a right sidebar with instructions and verification steps. The terminal at the bottom shows the execution of the 'npm run server' command and the output of the search results.

Langfuse tracing added for BM25 and hybrid search

The server now traces BM25 and hybrid search requests in the same style as the existing vector-search tracing.

What changed

- Added Langfuse trace creation for:
 - BM25 search requests
 - Hybrid search requests
- Added dedicated spans for:
 - BM25 search execution
 - Vector search execution within hybrid flow
 - Embedding generation
- Added trace updates and flushes on both success and error paths so the data is sent to Langfuse Cloud consistently.

Updated files

- index.js
- index_original.js

Verification

I verified the server files parse successfully with:

- node --check index.js
- node --check index_original.js

Both completed without syntax errors.

To see the traces in Langfuse, restart the server with your Langfuse credentials configured in the environment and

Describe what to build

Agent Auto

Local Default approvals (sandboxed)

Terminal Output:

```
[0] "createdAt": 1,
[0] "lastModifiedDate": 1,
[0] "risk": 1,
[0] "version": 1,
[0] "type": 1,
[0] "sourceFile": 1,
[0] "createdAt": 1,
[0] "score": 1
[0] }
[0] }
[0] }
[0] Found results: 5
[0] Sending response with 5 results
[0] Langfuse trace flushed: 42c249ee-7d88-4a21-86d9-c1a45c1920ac
^C[0] npm run server exited with code SIGINT
[1] npm run client exited with code SIGINT
ajayuchiha@Ajays-MacBook-Air rag-mongo-demo-v10 %
```

2.2 Terminal Log for the Hybrid Implementation:

```
Problems Output Debug Console Terminal Ports
[0] 'Radiology',
[0] 'Registration',
[0] 'SMS',
[0] 'Ward-Nursing',
[0] 'Wards-Doctors',
[0] 'Wards-MRD'
[0] ],
[0] priorities: [ 'P1', 'P2', 'P3', 'P4' ],
[0] risks: [ 'Critical', 'High', 'Low', 'Medium' ],
[0] types: [ 'Automated', 'Manual' ]
[0] }
[0] Hybrid Search request: "Share Diagnostic Reports with Patients via WhatsApp"
[0] BM25 Weight: 0.5, Vector Weight: 0.5
[0] Running BM25 search...
[0] Running vector search with Mistral AI...
[0] Combining results...
[0] Hybrid Search complete: 10 results in 1694ms
[0] Langfuse trace flushed: 2a1124b3-3d1a-4fff-9e4b-3887c819c218
[0]
```

2.3 Generation: (Hybrid)

cloud.langfuse.com/project/cms1qmlc20sbpad0h2kr4dacf/traces?peek=85eb3a9d-3c60-4b06-913e-d0ee209f3c5d&observation=748fec7a-b62e-4744-982a...

v4.4.0 Ajay's Organization Hobby First_llm_project

Trace search: 2a1124b3-3d1a-4fff-9e4b-3887c819c218

generate-embedding :

+ Add to datasets Annotate Playground

Add comment

2026-08-05 16:55:28.920

Latency: 0.94s Env: default 0 prompt → 0 completion (Σ 14)

mistral-embed

Preview Scores Log View Formatted JSON

"Share Diagnostic Reports with Patients via WhatsApp"

Output

{ 2 Items
tokens: 14
cost: 0.0000014000000000000000 }

Corrected Output

Click to add corrected output

Metadata

{ 5 Items
weights: { 2 Items
bm25: 0.5
vector: 0.5
}
timing: { 3 Items
bm25Time: 90
vectorTime: 1593
totalTime: 1694
}
stats: { 5 Items
foundInBoth: 5
foundInBm25Only: 1
foundInVectorOnly: 4
bm25ResultCount: 30
vectorResultCount: 30
}
resultCount: 10
searchType: "hybrid"
}

Tracing

Search - e.g. level:ERROR, -env:dev, latency:>2, scores.accuracy

Quality Slow Cost My Views 0 Table

Filters

Environment

Select Text

default 13

See docs on how to add environments to your observations-events.

API Key

Select Text

pk-ll-6350679... 13

Type

Select Text

SPAN 10

GENERATI... 3

Is Root Observation

True 4

False 9

Trace Name

Name

Select Text

search 4

bm25-search 3

Count

5

2

20:30

Start Time Type Name

2026-08-05 16:55:29 vector

2026-08-05 16:55:29 vector

2026-08-05 16:55:28 gener

2026-08-05 16:55:28 gener

2026-08-05 16:55:28 bm25

2026-08-05 16:55:28 bm25

2026-08-05 16:55:25 search

2026-08-05 16:55:25 search

2026-08-05 16:41:27 bm25

2026-08-05 16:41:24 search

2026-08-05 15:56:33 vector

2026-08-05 15:56:32 gener

2026-08-05 15:56:29 search

Graph

2.4: BM25:

cloud.langfuse.com/project/cms1qmlc20sbpad0h2kr4dacf/traces?peek=066fe946-c5ad-4016-81a7-99fd4c948ac9&observation=066fe946-c5ad-4016-81a7-...

v4.4.0

Ajay's Organization Hobby / First_llm_project

Tracing

Search - e.g. level:ERROR, -env:dev, latency:>2, scores.accuracy:>0.8

Quality Slow Cost My Views 0 Table Chart

Filters

Environment

API Key

Type

Is Root Observation

Trace Name

Name

search

Count

Start Time	Type	Name	Trace
2026-08-05 16:55:29	vector-search	search	search
2026-08-05 16:55:29	vector-search	search	search
2026-08-05 16:55:28	generate-embedding	search	search
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2026-08-05 16:55:28	bm25-search	search	search
2026-08-05 16:55:25	search	search	search
2026-08-05 16:55:25	search	search	search
2026-08-05 16:41:27	bm25-search	search	search
2026-08-05 16:41:24	search	search	search
2026-08-05 15:56:33	vector-search	search	search
2026-08-05 15:56:32	generate-embedding	search	search
2026-08-05 15:56:29	search	search	search

Trace search: a65473eb-b10d-4808-9cbb-cc26...

search 3.64s

bm25-search 0.12s

bm25-search

Add to datasets Annotate

Add comment

2026-08-05 16:41:27.808

Latency: 0.12s Env: default

Preview Scores Log View Formatted JSON Beta

```
5: 7
6: 165
7: 179
8: 97
9: 104
10: 17
11: 78
}
module: "AHC"
id: "TC_2736"
title: "AHCSummary ->AHCSummary ->Diagnostic & Imaging report"
description: "Verify Diagnostic & Imaging report"
steps: "1.Launch the application
2.Login to the application with valid credentials.
3.Click on Patient Service Tab and select AHC from drop down.
4.Navigate to Functions -> AHCSummary ->AHCSummary.
5.Select date or Enter UHID/AHC No and click on "Go" button.
6.Click on AHC Number.
7.Click on Diagnostic & Imaging link
8.Observe the results
9.Click on Download or print icon"
expectedResults: "Diagnostic & Imaging details should be displayed"
automationManual: "Automated"
priority: "P3"
risk: "Low"
createdAt: { @ Items }
}
score: 16.07637596130371
1: { 12 Items }
_id: { 1 Items }
buffer: { 12 Items }
a: 106
```